Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization

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***Abstract*— In this paper, we propose a Hessian-inverse-free prediction-correction method for time-varying strongly convex optimization with possibly nearly singular or highly ill-conditioned Hessians. The method couples a primal prediction-correction flow with an auxiliary matrix flow that generates inverse Hessian information through prediction, correction, and anchoring terms, thereby avoiding explicit matrix inversion. We establish positive definiteness and uniform boundedness of the auxiliary matrix state and derive an optimizer-tracking bound. Under explicit parameter conditions, the method achieves zero asymptotic tracking error without anchoring, while anchoring regulates the auxiliary matrix spectrum and yields a bounded, generally nonzero asymptotic tracking error. A numerical example illustrates the method near singularity.**

I. Introduction

Time-varying optimization arises naturally in many areas such as control, signal processing and robotics, where the objective function evolves continuously and the optimizer drifts with time. In such settings, the solution cannot be computed once offline and reused thereafter. Instead, it must be tracked online as the problem data change. This online tracking requirement is central to a broad range of applications, including adaptive traffic control [1], power grid optimization [2], decentralized learning [3] and target tracking [4]. The resulting challenge is to balance tracking accuracy, computational efficiency, and robustness to temporal variation.

A basic continuous-time approach is to use correction-only gradient flow, which updates the state using only instantaneous objective information and is attractive for its conceptual simplicity and low per-step complexity [5]. More broadly, continuous-time gradient, primal-dual, and projected primal-dual flows have also been developed for distributed and constrained time-varying optimization problems [6]–[8]. These flows show that real-time tracking of a moving optimizer is possible in a variety of settings. However, since they rely only on current-time optimality information, they typically converge to a bounded neighborhood of the optimal trajectory rather than achieving unbiased asymptotic tracking, so a nonzero steady-state tracking bias generally remains, even when strong convexity is imposed.

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To overcome this limitation, a natural idea is to explicitly exploit the temporal variation of the problem, which leads to the prediction-correction framework. In continuous time, prediction-correction flows combine a prediction term that compensates for the drift of the optimality conditions with a correction term that enforces instantaneous optimality [9]–[13]. Under suitable regularity conditions, such flows can achieve unbiased asymptotic tracking in the uniformly strongly convex setting. The same viewpoint has also inspired a broad class of discrete-time algorithms, where prediction-correction has become a standard paradigm for improving tracking accuracy in time-varying optimization [14]–[18]. In this context, Newton prediction-correction (Newton–PC) methods are particularly attractive, as they achieve unbiased asymptotic tracking in continuous time and can further improve discrete-time tracking accuracy through the use of second-order information [9]–[11].

Despite the advantages of Newton–PC methods, their continuous-time implementations typically rely on the inverse Hessian. This requirement is restrictive for two reasons. First, when the Hessian is nearly singular or highly ill-conditioned, its inverse can have an extremely large spectral norm and amplify numerical errors. Second, computing or applying the inverse Hessian generally requires repeated matrix inversion or factorization, which is more expensive and less amenable to efficient hardware implementation than basic matrix operations such as matrix multiplication. These issues make the classical Newton–PC flow less attractive in real-time applications involving nearly singular or highly ill-conditioned Hessians.

Meanwhile, a related but largely separate line of research has emerged in the Zhang Neural Network (ZNN) literature [19]–[21], where inverse-tracking dynamics are constructed for time-varying matrix inversion and online optimization. These works show that time-varying inverse information can be generated dynamically, rather than recomputed by explicit matrix inversion at each instant, and have demonstrated strong real-time and predictive capabilities in time-varying matrix problems. The underlying time-varying matrix inverse dynamics are well established. However, their systematic integration with a Newton–PC flow, together with an analysis of the resulting coupled optimization dynamics and the effect of anchoring on spectral boundedness and tracking accuracy, has received less attention.

Motivated by the prediction-correction framework in time-varying optimization and the inverse-tracking ideas in ZNN, this paper proposes a Hessian-inverse-free prediction-correction method for time-varying strongly convex op-

timization. The method augments the primal prediction-correction flow with an auxiliary matrix flow that dynamically generates inverse Hessian information, thereby avoiding explicit Hessian inversion while preserving the core prediction-correction structure. Our focus is on problems whose Hessians remain positive definite but can be nearly singular or highly ill-conditioned. The main contributions of this paper are summarized as follows.

- We propose a coupled prediction-correction method that integrates a primal Newton–PC flow with an auxiliary dynamic inverse-tracking flow. The auxiliary flow combines prediction, correction, and anchoring terms to generate inverse Hessian information online without explicit matrix inversion. We also clarify the role of each term.
- We establish structural guarantees for the auxiliary flow. In particular, the auxiliary matrix state remains positive definite and uniformly bounded. We also derive explicit spectral and inverse-tracking bounds.
- We derive an explicit optimizer-tracking bound that quantifies how the inverse-tracking bias affects optimization performance. The method achieves exact asymptotic tracking without anchoring under a suitable parameter condition and practical tracking with anchoring under a sufficient convergence condition.

The remainder of the paper is organized as follows. Section II introduces notation, mathematical preliminaries, and the problem statement. Section III presents the proposed Hessian-inverse-free prediction-correction flow and develops the main convergence analysis. Section IV reports numerical experiments, and Section V concludes the paper.

II. Preliminaries and Problem Statement

A. Notation

Let $\mathbb{R}$ and $\mathbb{R}_+$ denote the sets of real and nonnegative real numbers, respectively. For $n \geq 1$, the space of n-dimensional real vectors is denoted by $\mathbb{R}^n$, the space of $n\times n$ real symmetric matrices by $\mathbb{S}^n$, and the set of positive definite matrices in $\mathbb{S}^n$ by $\mathbb{S}^n_{++}$. The symbols $\boldsymbol{I}_n$ and $\mathbf{1}_n$ denote the $n \times n$ identity matrix and the all-ones vector in $\mathbb{R}^n$, respectively. For symmetric matrices $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{S}^n$, the notation $\boldsymbol{X} \preceq \boldsymbol{Y}$ means that $\boldsymbol{Y} - \boldsymbol{X}$ is positive semidefinite. For matrices, $\|\cdot\|_2$ and $\|\cdot\|_F$ denote the spectral norm and Frobenius norm, respectively, and $\langle \boldsymbol{M}, \boldsymbol{N}\rangle_F := \operatorname{tr}(\boldsymbol{M}^\top \boldsymbol{N})$ denotes the Frobenius inner product. For a symmetric matrix $\boldsymbol{X} \in \mathbb{S}^n$, $\lambda_{\min}(\boldsymbol{X})$ and $\lambda_{\max}(\boldsymbol{X})$ denote its smallest and largest eigenvalues. For vectors, $\|\cdot\|_2$ denotes the Euclidean norm. For a function $f : \mathbb{R}^n \times \mathbb{R}_+ \to \mathbb{R}$, we use $\nabla_{\boldsymbol{x}} f(\boldsymbol{x}, t)$ and $\nabla_{\boldsymbol{xx}} f(\boldsymbol{x}, t)$ to denote its gradient and Hessian with respect to $\boldsymbol{x}$, and use $\nabla_{\boldsymbol{x}t} f(\boldsymbol{x}, t)$, $\nabla_{\boldsymbol{xxx}} f(\boldsymbol{x}, t)$, and $\nabla_{\boldsymbol{xx}t} f(\boldsymbol{x}, t)$ to denote the mixed partial and higher-order derivatives used in the analysis. For a scalar-, vector-, or matrix-valued function $\xi(t)$, its total derivative with respect to t is denoted by $\dot{\xi}(t)$. When $\xi(t)$ is real-valued, its upper right Dini derivative is denoted by $D^+\xi(t) := \limsup_{h\to 0^+} \frac{\xi(t+h)-\xi(t)}{h}$.

B. Mathematical preliminaries

In this subsection, we recall two lemmas that will be repeatedly used in the subsequent analysis.

Lemma 1 (Young's inequality): For any $a, b \in \mathbb{R}$ and $\varepsilon > 0$,

$$ab \leq \frac{\varepsilon}{2}a^2 + \frac{1}{2\varepsilon}b^2.$$

Lemma 2: Let $V : [t_0, \infty) \to \mathbb{R}_+$ be differentiable. If there exist constants $c > 0$ and $d \geq 0$ such that

$$\dot{V}(t) \leq -cV(t) + d, \quad \forall\, t \geq t_0,$$

then

$$\limsup_{t\to\infty} V(t) \leq \frac{d}{c}.$$

Proof: By the comparison lemma in [22], we have

$$V(t) \leq e^{-c(t-t_0)}V(t_0) + \frac{d}{c}\big(1 - e^{-c(t-t_0)}\big), \quad \forall\, t \geq t_0.$$

Letting $t \to \infty$ yields the conclusion. ∎

C. Problem Statement

We consider the time-varying strongly convex optimization problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}, t), \tag{1}$$

where $f(\boldsymbol{x}, t)$ is sufficiently smooth in $(\boldsymbol{x}, t)$ so that the derivatives used below are well defined, in particular $\nabla_{\boldsymbol{xx}} f(\boldsymbol{x}, t)$ and $\nabla_{\boldsymbol{x}t} f(\boldsymbol{x}, t)$.

For the subsequent analysis, we impose the following assumption on the Hessian of the objective.

Assumption 1: There exist constants $0 < m \leq L$ such that, for all $\boldsymbol{x} \in \mathbb{R}^n$ and $t \geq 0$,

$$m\boldsymbol{I}_n \preceq \nabla_{\boldsymbol{xx}} f(\boldsymbol{x}, t) \preceq L\boldsymbol{I}_n.$$

Remark 1: Assumption 1 imposes uniform spectral bounds on the Hessian of $f(\boldsymbol{x}, t)$. In particular, for each $t \geq 0$, the function $f(\cdot, t)$ is uniformly m-strongly convex and L-smooth, and problem (1) admits a unique optimizer $\boldsymbol{x}^\star(t)$. This assumption is standard in time-varying optimization. See, e.g., [9], [14], [16]–[18]. Here $m > 0$ is allowed to be very small, so the framework includes strongly convex problems whose Hessians are nearly singular or highly ill-conditioned, which is the setting of primary interest in this paper.

III. Method Design and Convergence Analysis

To solve (1), we propose the following $\boldsymbol{x}$–$\boldsymbol{H}$ coupled flow

$$\dot{\boldsymbol{H}} = -\boldsymbol{H}\dot{\boldsymbol{A}}\boldsymbol{H} + \alpha\boldsymbol{H}(\boldsymbol{I}_n - \boldsymbol{A}\boldsymbol{H}) + \gamma\boldsymbol{H}(\eta\boldsymbol{I}_n - \boldsymbol{H}), \tag{2a}$$

$$\dot{\boldsymbol{x}} = -\boldsymbol{H}(\beta\boldsymbol{g} + \boldsymbol{b}). \tag{2b}$$

For the solution trajectory $(\boldsymbol{x}(t), \boldsymbol{H}(t))$ of (2a)–(2b), we use the shorthand $\boldsymbol{g} := \nabla_{\boldsymbol{x}} f(\boldsymbol{x}(t), t)$, $\boldsymbol{A} := \nabla_{\boldsymbol{xx}} f(\boldsymbol{x}(t), t)$, and $\boldsymbol{b} := \nabla_{\boldsymbol{x}t} f(\boldsymbol{x}(t), t)$. In (2a), $\dot{\boldsymbol{A}}$ denotes the total derivative

of $\boldsymbol{A}$ along the trajectory. When $\nabla_{\boldsymbol{xxx}} f$ and $\nabla_{\boldsymbol{xxt}} f$ exist, it can be expressed as $\dot{\boldsymbol{A}} = \sum_{i=1}^{n} \nabla_{\boldsymbol{xx}x_i} f(\boldsymbol{x}(t), t)\dot{x}_i + \nabla_{\boldsymbol{xxt}} f(\boldsymbol{x}(t), t)$. Moreover, $\alpha > 0$, $\beta > 0$, $\gamma \geq 0$, and $\eta > 0$ are design parameters subject to the theoretical conditions derived later.

Remark 2: The coupled $\boldsymbol{x}$–$\boldsymbol{H}$ flow has a unified prediction-correction interpretation. In the $\boldsymbol{H}$ flow, $-\boldsymbol{H}\dot{\boldsymbol{A}}\boldsymbol{H}$ is motivated by differentiating $\boldsymbol{AH} = \boldsymbol{I}_n$ under the ideal relation $\boldsymbol{H} = \boldsymbol{A}^{-1}$ and predicts the variation of inverse Hessian information, as in ZNN dynamics for time-varying matrix inversion [19]–[21]. The term $\alpha\boldsymbol{H}(\boldsymbol{I}_n - \boldsymbol{AH})$ corrects the inverse residual, while $\gamma\boldsymbol{H}(\eta\boldsymbol{I}_n - \boldsymbol{H})$ anchors the spectral scale of $\boldsymbol{H}$. In the $\boldsymbol{x}$ flow, $-\boldsymbol{Hb}$ predicts optimizer motion and $-\beta\boldsymbol{Hg}$ corrects the optimality residual.

Remark 3: For numerical simulation, the continuous-time flow (2a)–(2b) can be discretized by the forward Euler method with sampling period $h = t_{k+1} - t_k$. In practice, derivatives with respect to time are often unavailable in closed form, so finite-difference approximations may be used. In particular, the mixed derivative term $\boldsymbol{b}(t_k) = \nabla_{\boldsymbol{x}t} f(\boldsymbol{x}(t_k), t_k)$ can be approximated by $\boldsymbol{b}(t_k) \approx (\nabla_{\boldsymbol{x}} f(\boldsymbol{x}_k, t_k) - \nabla_{\boldsymbol{x}} f(\boldsymbol{x}_k, t_{k-1}))/h$, as in [14]. Since $\dot{\boldsymbol{A}}$ is the total Hessian derivative along the solution trajectory, it can be approximated by the first-order backward difference $\dot{\boldsymbol{A}}(t_k) \approx (\boldsymbol{A}(\boldsymbol{x}_k, t_k) - \boldsymbol{A}(\boldsymbol{x}_{k-1}, t_{k-1}))/h$. The former approximates partial-time variation at the current state, whereas the latter follows the numerical trajectory. A rigorous convergence analysis of the discretized flow is left for future work.

For the analysis of the proposed flow, we further impose the following trajectory-level regularity assumption.

Assumption 2: Along the solution trajectory $\{(\boldsymbol{x}(t), \boldsymbol{H}(t))\}$ of (2a)–(2b), there exist constants $M_b > 0$ and $M_A > 0$ such that

$$\|\boldsymbol{b}\|_2 \leq M_b, \quad \|\dot{\boldsymbol{A}}\|_2 \leq M_A.$$

Remark 4: Assumption 2 bounds the mixed temporal variation of the gradient and the total variation of the Hessian along the closed-loop trajectory. It is an additional trajectory-level condition for analyzing the Hessian-inverse-free design. Related bounds on higher-order derivatives appear in [14], [16], [17]. The condition is directly checkable for structured problems. For a time-varying quadratic problem, $\nabla_{\boldsymbol{xxx}} f = 0$ and $\dot{\boldsymbol{A}}$ is prescribed. More generally, it holds when the relevant higher-order derivatives and $\dot{\boldsymbol{x}}$ are bounded along the solution trajectory. For a general nonlinear problem, it remains a closed-loop regularity requirement.

We begin the analysis by establishing the basic structural properties of the $\boldsymbol{H}$ flow, as stated in Lemmas 3 and 4.

Lemma 3: Suppose that Assumptions 1 and 2 hold, and

$$\boldsymbol{H}(0) \in \mathbb{S}^n_{++}, \tag{3}$$

then $\boldsymbol{H}(t) \in \mathbb{S}^n_{++}$ for all $t \geq 0$.

Proof: We first show that $\boldsymbol{H}(t)$ remains in $\mathbb{S}^n$. For each $t \geq 0$, both $\boldsymbol{A}(t)$ and $\dot{\boldsymbol{A}}(t)$ are symmetric. Thus, whenever $\boldsymbol{H}(t) \in \mathbb{S}^n$, the right-hand side (RHS) of (2a) is also symmetric. This means that $\mathbb{S}^n$ is invariant under (2a). Since $\boldsymbol{H}(0) \in \mathbb{S}^n$, we obtain $\boldsymbol{H}(t) \in \mathbb{S}^n$ for all $t \geq 0$.

Since $\boldsymbol{H}(t) \in \mathbb{S}^n$ for all $t \geq 0$, proving $\boldsymbol{H}(t) \in \mathbb{S}^n_{++}$ is equivalent to showing that $\lambda_{\min}(\boldsymbol{H}(t)) > 0$ for all $t \geq 0$. For brevity, we write $\lambda_{\min}(t) := \lambda_{\min}(\boldsymbol{H}(t))$. By the variational characterization of the smallest eigenvalue [23], $\lambda_{\min}(t) = \min_{\|\boldsymbol{v}\|_2=1} \boldsymbol{v}^\top \boldsymbol{H}(t)\boldsymbol{v}$. When the smallest eigenvalue has multiplicity greater than one, $\lambda_{\min}(t)$ may fail to be differentiable. Therefore, instead of the ordinary derivative, we analyze its variation via the upper right Dini derivative [22]. By Danskin's theorem [24],

$$D^+\lambda_{\min}(t) = \min_{\boldsymbol{v} \in \mathcal{V}(t)} \boldsymbol{v}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{v}, \tag{4}$$

where $\mathcal{V}(t) := \operatorname{argmin}_{\|\boldsymbol{v}\|_2=1} \boldsymbol{v}^\top \boldsymbol{H}(t)\boldsymbol{v}$. Since $\boldsymbol{H}(t) \in \mathbb{S}^n$, any minimizer $\boldsymbol{v} \in \mathcal{V}(t)$ is a unit eigenvector corresponding to $\lambda_{\min}(t)$, i.e., $\boldsymbol{H}(t)\boldsymbol{v} = \lambda_{\min}(t)\boldsymbol{v}$. Substituting (2a) gives

$$\begin{aligned}\boldsymbol{v}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{v} = &-\lambda^2_{\min}(t)\boldsymbol{v}^\top \dot{\boldsymbol{A}}(t)\boldsymbol{v} + \alpha\lambda_{\min}(t) \\ &- \alpha\lambda^2_{\min}(t)\boldsymbol{v}^\top \boldsymbol{A}(t)\boldsymbol{v} + \gamma\big(\eta\lambda_{\min}(t) - \lambda^2_{\min}(t)\big).\end{aligned}$$

By Assumptions 1–2, we have $\boldsymbol{v}^\top \boldsymbol{A}(t)\boldsymbol{v} \leq L$ and $|\boldsymbol{v}^\top \dot{\boldsymbol{A}}(t)\boldsymbol{v}| \leq \|\dot{\boldsymbol{A}}(t)\|_2 \leq M_A$. Therefore,

$$\boldsymbol{v}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{v} \geq (\alpha + \gamma\eta)\lambda_{\min}(t) - \big(M_A + \alpha L + \gamma\big)\lambda^2_{\min}(t).$$

Since the above inequality holds for every $\boldsymbol{v} \in \mathcal{V}(t)$, it also holds for $\min_{\boldsymbol{v} \in \mathcal{V}(t)} \boldsymbol{v}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{v}$. Combining this with (4) gives

$$D^+\lambda_{\min}(t) \geq (\alpha + \gamma\eta)\lambda_{\min}(t) - \big(M_A + \alpha L + \gamma\big)\lambda^2_{\min}(t). \tag{5}$$

Since (5) is not a classical ordinary differential equation (ODE), we introduce the following auxiliary continuously differentiable scalar ODE:

$$\dot{y}(t) = (\alpha + \gamma\eta)y(t) - \big(M_A + \alpha L + \gamma\big)y^2(t), \tag{6}$$

with $y(0) = \lambda_{\min}(0) > 0$. Since the RHS of (6) is locally Lipschitz, (6) admits a unique classical solution (see [25], Corollary 1). By the comparison principle for upper right Dini derivatives [22], we have $\lambda_{\min}(t) \geq y(t)$ for all $t \geq 0$. Therefore, it remains to show that $y(t) > 0$ for all $t \geq 0$. Noting that $y \equiv 0$ is also a solution, uniqueness of classical solutions implies that the solution with $y(0) > 0$ cannot intersect $y \equiv 0$. Hence, $y(t) > 0$ for all $t \geq 0$. Therefore, $\lambda_{\min}(t) > 0$ for all $t \geq 0$, which proves $\boldsymbol{H}(t) \in \mathbb{S}^n_{++}$ for all $t \geq 0$. ∎

Lemma 4: Suppose that Assumptions 1–2 and (3) hold, and

$$\alpha m + \gamma > M_A, \tag{7}$$

then

$$\|\boldsymbol{H}(t)\|_2 \leq H_1, \quad \forall\, t \geq 0,$$

where

$$H_1 := \max\left\{\lambda_{\max}(\boldsymbol{H}(0)), \frac{\alpha + \gamma\eta}{\alpha m + \gamma - M_A}\right\}.$$

Proof: Lemma 3 ensures that $\boldsymbol{H}(t) \in \mathbb{S}^n_{++}$ for all $t \geq 0$, and hence its spectral norm coincides with its largest

eigenvalue. Therefore, establishing a uniform upper bound on $\|\boldsymbol{H}(t)\|_2$ is equivalent to deriving a uniform upper bound for $\lambda_{\max}(\boldsymbol{H}(t))$. For brevity, we write $\lambda_{\max}(t) := \lambda_{\max}(\boldsymbol{H}(t))$. Then, as in the proof of Lemma 3, the variational characterization of the largest eigenvalue together with Danskin's theorem gives

$$D^+\lambda_{\max}(t) = \max_{\boldsymbol{w}\in\mathcal{W}(t)} \boldsymbol{w}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{w}, \tag{8}$$

where $\mathcal{W}(t) := \operatorname{argmax}_{\|\boldsymbol{w}\|_2=1} \boldsymbol{w}^\top \boldsymbol{H}(t)\boldsymbol{w}$. Moreover, for any $\boldsymbol{w} \in \mathcal{W}(t)$, one has $\boldsymbol{H}(t)\boldsymbol{w} = \lambda_{\max}(t)\boldsymbol{w}$. Substituting (2a) gives

$$\begin{aligned}\boldsymbol{w}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{w} &= -\lambda_{\max}^2(t)\boldsymbol{w}^\top \dot{\boldsymbol{A}}(t)\boldsymbol{w} + \alpha\lambda_{\max}(t) \\ &\quad - \alpha\lambda_{\max}^2(t)\boldsymbol{w}^\top \boldsymbol{A}(t)\boldsymbol{w} + \gamma\big(\eta\lambda_{\max}(t) - \lambda_{\max}^2(t)\big).\end{aligned}$$

By Assumptions 1–2, we have $\boldsymbol{w}^\top \boldsymbol{A}(t)\boldsymbol{w} \geq m$ and $|\boldsymbol{w}^\top \dot{\boldsymbol{A}}(t)\boldsymbol{w}| \leq \|\dot{\boldsymbol{A}}(t)\|_2 \leq M_A$. Therefore,

$$\boldsymbol{w}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{w} \leq (\alpha+\gamma\eta)\lambda_{\max}(t) - \big(\alpha m+\gamma-M_A\big)\lambda_{\max}^2(t).$$

Since the above inequality holds for every $\boldsymbol{w} \in \mathcal{W}(t)$, it also holds for $\max_{\boldsymbol{w}\in\mathcal{W}(t)} \boldsymbol{w}^\top \dot{\boldsymbol{H}}(t)\boldsymbol{w}$. Combining this with (8) gives

$$D^+\lambda_{\max}(t) \leq (\alpha+\gamma\eta)\lambda_{\max}(t) - \big(\alpha m+\gamma-M_A\big)\lambda_{\max}^2(t).$$

To bound $\lambda_{\max}(t)$, consider the scalar ODE:

$$\dot{z}(t) = (\alpha+\gamma\eta)z(t) - \big(\alpha m+\gamma-M_A\big)z^2(t), \tag{9}$$

with $z(0) = \lambda_{\max}(0)$. Since the RHS of (9) is locally Lipschitz, (9) admits a unique classical solution. Moreover, by the comparison principle for upper right Dini derivatives [22], we have $\lambda_{\max}(t) \leq z(t)$ for all $t \geq 0$. Therefore, it remains to show that $z(t)$ admits a uniform upper bound for all $t \geq 0$.

Under (7), we have $\alpha m + \gamma - M_A > 0$, and hence

$$z_\star := \frac{\alpha+\gamma\eta}{\alpha m+\gamma-M_A}$$

is a positive equilibrium of (9). Since $z(0) = \lambda_{\max}(\boldsymbol{H}(0))$, the constant H_1 in Lemma 4 satisfies $H_1 = \max\{z(0), z_\star\}$. We consider the following two cases:

- If $z(0) \leq z_\star$, then $H_1 = z_\star$. Since $z \equiv z_\star$ is also a solution of (9), uniqueness of classical solutions implies that $z(t)$ cannot intersect $z_\star$. Hence, $z(t) \leq z_\star = H_1$ for all $t \geq 0$.
- If $z(0) > z_\star$, then $H_1 = z(0)$. Moreover, $\dot{z}(t) < 0$ whenever $z(t) > z_\star$. Therefore, $z(t)$ cannot exceed its initial value, and thus $z(t) \leq z(0) = H_1$ for all $t \geq 0$.

In both cases, $z(t) \leq H_1$ for all $t \geq 0$. It follows that $\|\boldsymbol{H}(t)\|_2 = \lambda_{\max}(t) \leq z(t) \leq H_1$ for all $t \geq 0$. ∎

Lemmas 3–4 ensure that $\boldsymbol{H}(t)$ remains positive definite and uniformly bounded for all $t \geq 0$ and that $\|\boldsymbol{H}(t)\|_2$ is uniformly bounded. We thus introduce the auxiliary matrix flow $\boldsymbol{P} := \boldsymbol{H}^{-1}$. Differentiating $\boldsymbol{P}(t)\boldsymbol{H}(t) = \boldsymbol{I}_n$ and substituting it into (2a) yields

$$\dot{\boldsymbol{P}} = \dot{\boldsymbol{A}} - \alpha(\boldsymbol{P}-\boldsymbol{A}) - \gamma(\eta\boldsymbol{P}-\boldsymbol{I}_n), \tag{10}$$

where $\boldsymbol{P}(0) = \boldsymbol{H}^{-1}(0)$.

Remark 5: The auxiliary $\boldsymbol{P}$ flow is introduced to simplify the analysis. Compared with the nonlinear $\boldsymbol{H}$ flow in (2a), the dynamics in (10) is linear in $\boldsymbol{P}$, which makes the subsequent convergence analysis more transparent. The proposed method is still given by the coupled $\boldsymbol{x}$–$\boldsymbol{H}$ flow in (2a)–(2b). Neither $\boldsymbol{H}^{-1}$ nor $\boldsymbol{A}^{-1}$ needs to be computed in its implementation.

Next, we will analyze the asymptotic tracking performance of $\boldsymbol{P}(t)$ with respect to the time-varying Hessian matrix $\boldsymbol{A}(t)$. To this end, define $\boldsymbol{E} := \boldsymbol{P} - \boldsymbol{A}$. The following lemma gives an asymptotic bound on $\|\boldsymbol{E}\|_F$.

Lemma 5: Suppose that Assumptions 1–2 and (3) hold. Then

$$\limsup_{t\to\infty} \|\boldsymbol{E}\|_F \leq \frac{\gamma\rho}{\alpha+\gamma\eta}, \tag{11}$$

where $\rho := \sqrt{n}\max\{|1-\eta m|, |1-\eta L|\}$.

Proof: From (10) and the definition $\boldsymbol{E} := \boldsymbol{P} - \boldsymbol{A}$, we have

$$\dot{\boldsymbol{E}} = -(\alpha+\gamma\eta)\boldsymbol{E} + \gamma(\boldsymbol{I}_n - \eta\boldsymbol{A}).$$

Consider the following Lyapunov candidate

$$V_E := \frac{1}{2}\|\boldsymbol{E}\|_F^2.$$

Then

$$\begin{aligned}\dot{V}_E &= \langle \boldsymbol{E}, \dot{\boldsymbol{E}}\rangle_F \\ &= -(\alpha+\gamma\eta)\|\boldsymbol{E}\|_F^2 + \gamma\langle \boldsymbol{E}, \boldsymbol{I}_n - \eta\boldsymbol{A}\rangle_F \\ &\leq -(\alpha+\gamma\eta)\|\boldsymbol{E}\|_F^2 + \gamma\|\boldsymbol{E}\|_F\|\boldsymbol{I}_n - \eta\boldsymbol{A}\|_F.\end{aligned}$$

Moreover, since $m\boldsymbol{I}_n \preceq \boldsymbol{A} \preceq L\boldsymbol{I}_n$, we have $\|\boldsymbol{I}_n - \eta\boldsymbol{A}\|_2 \leq \max\{|1-\eta m|, |1-\eta L|\}$, and thus $\|\boldsymbol{I}_n - \eta\boldsymbol{A}\|_F \leq \sqrt{n}\|\boldsymbol{I}_n - \eta\boldsymbol{A}\|_2 \leq \rho$. Therefore,

$$\dot{V}_E \leq -2(\alpha+\gamma\eta)V_E + \gamma\rho\sqrt{2V_E}. \tag{12}$$

Applying Lemma 1 with $a = \sqrt{2V_E}$, $b = \gamma\rho$, and $\varepsilon = \alpha+\gamma\eta$ gives

$$\gamma\rho\sqrt{2V_E} \leq (\alpha+\gamma\eta)V_E + \frac{(\gamma\rho)^2}{2(\alpha+\gamma\eta)}.$$

Substituting this into (12) yields

$$\dot{V}_E \leq -(\alpha+\gamma\eta)V_E + \frac{(\gamma\rho)^2}{2(\alpha+\gamma\eta)}.$$

Therefore, by Lemma 2,

$$\limsup_{t\to\infty} V_E(t) \leq \frac{(\gamma\rho)^2}{2(\alpha+\gamma\eta)^2}.$$

Since $2V_E = \|\boldsymbol{E}\|_F^2$, it follows that

$$\limsup_{t\to\infty} \|\boldsymbol{E}\|_F \leq \frac{\gamma\rho}{\alpha+\gamma\eta},$$

which proves (11). ∎

Remark 6: Lemma 5 characterizes the asymptotic bias in tracking the Hessian with $\boldsymbol{P} = \boldsymbol{H}^{-1}$. When $\gamma = 0$, $\limsup_{t\to\infty} \|\boldsymbol{E}(t)\|_F = 0$, so $\boldsymbol{H}^{-1}(t)$ tracks $\boldsymbol{A}(t)$ without bias. When $\gamma > 0$, the anchoring term generally produces a nonzero but explicitly bounded inverse-tracking bias.

Define $\boldsymbol{R} := \boldsymbol{EH} = \boldsymbol{I}_n - \boldsymbol{AH}$. Under Assumptions 1–2 together with conditions (3) and (7), Lemmas 4–5 imply

$$\begin{aligned}\limsup_{t\to\infty} \|\boldsymbol{R}\|_2 &\le \limsup_{t\to\infty} \|\boldsymbol{E}\|_2 \|\boldsymbol{H}\|_2 \\ &\le \limsup_{t\to\infty} \|\boldsymbol{E}\|_F \|\boldsymbol{H}\|_2 \\ &\le \frac{\gamma\rho}{\alpha+\gamma\eta} H_1 =: R_\infty,\end{aligned} \tag{13}$$

where the first inequality uses the submultiplicativity of the spectral norm, the second follows from $\|\cdot\|_2 \le \|\cdot\|_F$, and the last combines Lemmas 4 and 5. The bound R_∞ in (13) is then used to derive an optimizer-tracking error bound.

Theorem 1: Suppose that Assumptions 1–2 and conditions (3), (7) hold. If $R_\infty < 1$, then

$$\limsup_{t\to\infty} \|\boldsymbol{x}(t) - \boldsymbol{x}^\star(t)\|_2 \le \frac{R_\infty}{m\beta(1-R_\infty)} M_b. \tag{14}$$

Proof: Since $R_\infty < 1$, choose any sufficiently small $\delta > 0$ such that $r_\delta := R_\infty + \delta < 1$. By (13), there exists $T_\delta \ge 0$ such that

$$\|\boldsymbol{R}(t)\|_2 \le r_\delta, \quad \forall\, t \ge T_\delta.$$

Consider the following Lyapunov candidate

$$V_g := \frac{1}{2}\|\boldsymbol{g}\|_2^2.$$

From the $\boldsymbol{x}$ flow in (2b), and $\boldsymbol{R} = \boldsymbol{I}_n - \boldsymbol{AH}$, we have

$$\dot{\boldsymbol{g}} = \boldsymbol{A}\dot{\boldsymbol{x}} + \boldsymbol{b} = -\boldsymbol{AH}(\beta\boldsymbol{g} + \boldsymbol{b}) + \boldsymbol{b} = \beta(\boldsymbol{R} - \boldsymbol{I}_n)\boldsymbol{g} + \boldsymbol{Rb}.$$

Hence, for all $t \ge T_\delta$,

$$\begin{aligned}\dot{V}_g &= \langle \boldsymbol{g}, \dot{\boldsymbol{g}} \rangle \\ &= \beta \boldsymbol{g}^\top (\boldsymbol{R} - \boldsymbol{I}_n)\boldsymbol{g} + \boldsymbol{g}^\top \boldsymbol{Rb} \\ &\le -\beta(1-r_\delta)\|\boldsymbol{g}\|_2^2 + r_\delta M_b \|\boldsymbol{g}\|_2 \\ &= -2\beta(1-r_\delta) V_g + r_\delta M_b \sqrt{2V_g}.\end{aligned}$$

Applying Lemma 1 with $a = \sqrt{2V_g}$, $b = r_\delta M_b$, and $\varepsilon = \beta(1-r_\delta)$ yields $r_\delta M_b \sqrt{2V_g} \le \beta(1-r_\delta)V_g + (r_\delta M_b)^2/[2\beta(1-r_\delta)]$. Substituting this inequality into the above estimate gives

$$\dot{V}_g \le -\beta(1-r_\delta)V_g + \frac{(r_\delta M_b)^2}{2\beta(1-r_\delta)}, \quad \forall\, t \ge T_\delta.$$

Therefore, by Lemma 2,

$$\limsup_{t\to\infty} V_g(t) \le \frac{(r_\delta M_b)^2}{2\beta^2(1-r_\delta)^2}.$$

Since $2V_g = \|\boldsymbol{g}\|_2^2$, it follows that

$$\limsup_{t\to\infty} \|\boldsymbol{g}\|_2 \le \frac{r_\delta}{\beta(1-r_\delta)} M_b.$$

Finally, letting $\delta \to 0^+$ yields $r_\delta \to R_\infty$ and

$$\limsup_{t\to\infty} \|\boldsymbol{g}\|_2 \le \frac{R_\infty}{\beta(1-R_\infty)} M_b. \tag{15}$$

Strong convexity gives $\|\boldsymbol{g}(t)\|_2 \ge m\|\boldsymbol{x}(t) - \boldsymbol{x}^\star(t)\|_2$. Combining this inequality with (15) yields (14):

$$\limsup_{t\to\infty} \|\boldsymbol{x}(t) - \boldsymbol{x}^\star(t)\|_2 \le \frac{R_\infty}{m\beta(1-R_\infty)} M_b.$$

This concludes the proof. ∎

Corollary 1: Suppose that Assumptions 1–2 and condition (3) hold. If $\gamma = 0$ and $\alpha m > M_A$, then $R_\infty = 0$, and $\limsup_{t\to\infty} \|\boldsymbol{x}(t) - \boldsymbol{x}^\star(t)\|_2 = 0$.

Remark 7: Without anchoring, $\alpha m > M_A$ eliminates the asymptotic inverse-tracking bias and yields exact asymptotic tracking by Corollary 1. When m is very small, however, this condition may require a large inverse-correction gain α. The anchoring term relaxes the boundedness condition to $\alpha m + \gamma > M_A$ and helps regulate the spectral scale of $\boldsymbol{H}(t)$. This benefit comes with a generally nonzero inverse-tracking bias. Provided additionally that $R_\infty < 1$, Theorem 1 gives an explicit practical tracking bound.

Remark 8: The sufficient convergence condition $R_\infty < 1$ can be checked from (13) and H_1. If $\lambda_{\max}(\boldsymbol{H}(0)) \le z_\star$, then $H_1 = z_\star$. A sufficient condition is $\gamma\rho/(\alpha m + \gamma - M_A) < 1$. The parameter β does not affect this condition and scales the bounds in (14) and (15) proportionally to β^{-1}.

IV. Numerical Experiments

We consider a nearly singular time-varying quadratic problem:

$$f(\boldsymbol{x}, t) = \frac{1}{2}(\boldsymbol{x} - \boldsymbol{c}(t))^\top \boldsymbol{A}(t)(\boldsymbol{x} - \boldsymbol{c}(t)), \tag{16}$$

where $\boldsymbol{x}, \boldsymbol{c}(t) \in \mathbb{R}^6$, $\boldsymbol{A}(t) \in \mathbb{R}^{6\times 6}$, and the optimal trajectory is $\boldsymbol{x}^\star(t) = \boldsymbol{c}(t)$. Here $\boldsymbol{c}_i(t) = c_i \sin(\omega_c t + \phi_i)$ with $\omega_c = 0.3$, and $\boldsymbol{A}(t) \in \mathbb{S}^6_{++}$ is constructed from a fixed orthogonal matrix and time-varying eigenvalues. In particular, $\boldsymbol{A}(t)$ is designed so that its smallest eigenvalue reaches 10^{-6}, and hence the strong convexity constant satisfies $m \approx 10^{-6}$. Therefore, $f(\cdot, t)$ remains strongly convex, but its Hessian is nearly singular and highly ill-conditioned, which allows the numerical test to focus on the main setting of interest in this paper.

We test the following three representative continuous-time methods on problem (16):

- the gradient flow: $\dot{\boldsymbol{x}} = -\beta\boldsymbol{g}$.
- the Newton–PC flow [9]: $\dot{\boldsymbol{x}} = -\boldsymbol{A}^{-1}(\beta\boldsymbol{g} + \boldsymbol{b})$.
- the proposed $\boldsymbol{x}$–$\boldsymbol{H}$ flow in (2a)–(2b).

All methods are initialized at $\boldsymbol{x}(0) = 3\,\mathbf{1}_6$ and simulated by the forward Euler method with stepsize $h = 10^{-4}$.

We first fix $\alpha = 5.0$, $\eta = 1$, and $\gamma = 0.1$, and test the proposed $\boldsymbol{x}$–$\boldsymbol{H}$ flow with $\beta \in \{1, 10\}$. For comparison, both the gradient flow and Newton–PC use $\beta = 1$. Fig. 1 shows that Newton–PC achieves the smallest error over much of the interval, but exhibits a clear increase when $\boldsymbol{A}(t)$ becomes highly ill-conditioned. In the ideal continuous-time model, Newton–PC achieves zero asymptotic tracking error under the stated regularity conditions. The observed spikes are therefore consistent with its numerical sensitivity when the Hessian is nearly singular. With $\gamma = 0.1$, the anchoring term

generally introduces an asymptotic tracking bias, while the proposed flow yields visibly smoother trajectories in this example. In addition, increasing β from 1 to 10 further improves the steady-state tracking accuracy in this example, reducing the error from about 0.05 to about 0.02.

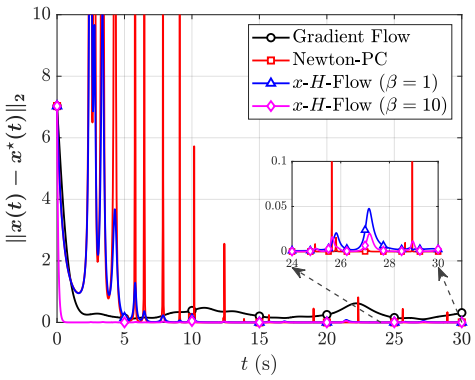


Fig. 1. Tracking error trajectories for the time-varying quadratic problem.

We next fix $\alpha = 5.0$, $\eta = 1$, and $\beta = 1$, and vary γ to examine the effect of the anchoring term on the inverse surrogate. Since $\boldsymbol{H}(t)$ plays a role analogous to $\boldsymbol{A}(t)^{-1}$ in Newton–PC, Table I compares their maximum spectral norms. The results show that $\|\boldsymbol{A}(t)^{-1}\|_2$ in Newton–PC becomes extremely large near singularity, whereas $\|\boldsymbol{H}(t)\|_2$ remains much smaller and decreases further as γ increases. This indicates that the anchoring term effectively limits the growth of $\boldsymbol{H}(t)$.

TABLE I
MAXIMUM SPECTRAL NORMS IN NEWTON–PC AND THE PROPOSED FLOW.

Method	**Quantity**	**Max. norm**
Newton–PC	$\|\boldsymbol{A}^{-1}(t)\|_2$	9.99×10^5
$\boldsymbol{x}$–$\boldsymbol{H}$, $\gamma = 0.05$	$\|\boldsymbol{H}(t)\|_2$	1.15×10^2
$\boldsymbol{x}$–$\boldsymbol{H}$, $\gamma = 0.10$	$\|\boldsymbol{H}(t)\|_2$	5.81×10^1
$\boldsymbol{x}$–$\boldsymbol{H}$, $\gamma = 0.20$	$\|\boldsymbol{H}(t)\|_2$	2.95×10^1

V. CONCLUSION

This paper developed a Hessian-inverse-free prediction-correction method and clarified the trade-off between tracking accuracy and spectral regulation near singularity. Future work will consider general convex problems without uniform strong convexity and discretization schemes for the proposed flow.

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